

**Never Miss a Dose: A Technology-Based Solution
for Medication Adherence
Among Older Adults**

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I. Introduction

At present, many older adults are required to take different medications to manage their health conditions and maintain their daily health. While taking medication as prescribed is important, managing several medicines at different times can become difficult, especially for older adults who may experience forgetfulness, complicated schedules, confusing instructions, or limited assistance from family members and caregivers. Throughout their daily lives, older adults may struggle to remember the correct medicine, dosage, and time for taking their medication. In this sense, medication non-adherence can result in missed doses, incorrect doses, or medication being taken at the wrong time, creating problems that can negatively affect the health and well-being of older adults.

Medication non-adherence affects approximately 22% to 68% of older adults, showing that failure to follow prescribed medication instructions is a significant concern among this population: complex medication schedules, memory difficulties, multiple medications, and medication costs can contribute to this problem. According to Stanly, Vilakkathala, and George (2025), medication non-adherence among older adults is associated with factors such as forgetfulness, polypharmacy, treatment complexity, physical difficulties, and limited social support.

This particular problem of medication non-adherence among older adults relates to the modern healthcare problem of ensuring that individuals take their medications correctly and consistently. It is important to address this problem by providing an ideal solution that can help older adults manage their medications while also allowing caregivers to monitor their medication-taking activities. In this proposal, It seeks to detail and scope the problem of medication non-adherence among older adults and to outline an ideal technological solution through an automated medication reminder and monitoring application called **Medi-Remind**.

Problem description

In modern times, medication non-adherence can occur for different reasons, but it can be particularly challenging for older adults who take multiple medications for different health conditions. Older adults may need to remember several medication

names, dosages, and schedules throughout the day. When these responsibilities become difficult to manage, individuals may forget to take their medication, take the wrong dosage, or take their medication at the wrong time. According to Ge, Heng, and Yap (2023), medication non-adherence affects approximately 22% to 68% of older adults, highlighting the importance of providing practical methods to help this population manage their medications.

One of the major problems associated with medication non-adherence is forgetfulness. Older adults may forget to take their medication because of interruptions in their daily routines or because they have several medications that must be taken at different times. Managing multiple medications can also create confusion because each medicine may have a different dosage and schedule. Stanly, Vilakkathala, and George (2025) explain that forgetfulness, polypharmacy, complex treatment regimens, physical difficulties, and limited social support can contribute to medication non-adherence among older adults.

Another problem is the complexity of medication schedules. Some medications may need to be taken once a day, while others may need to be taken several times a day or at specific times. This can make it difficult for older adults to remember every medication they need to take. Medication costs may also create another barrier, as some individuals may have difficulty obtaining their prescribed medications consistently. With this in mind, medication non-adherence can result in poor disease control, treatment failure, increased health complications, and additional healthcare needs.

In addition, caregivers may have difficulty monitoring medication intake when they are not physically present with the older adult. Traditional methods such as written schedules, calendars, or ordinary alarms may remind a person that medication is due, but they may not provide information about whether the medication was actually taken. Because of this, there is a need for a system that can provide reminders while also recording medication-taking activities.

With this observation in mind, it is worth noting how medication non-adherence can affect the lives of older adults and their caregivers. However, traditional medication reminder methods do not always provide automated monitoring, medication history, or caregiver notifications. Some solutions are limited to reminding users about their

medication but do not allow them to confirm whether the dose was taken or missed. These limitations show the need for a more organized and accessible technology-based solution.

With this in mind, the proposal of an ideal solution can be addressed through an automated medication reminder and monitoring application that can help older adults manage their medication schedules properly while providing caregivers with additional support.

Proposal Solution

The proposed solution to mitigate medication non-adherence among older adults is the development of an automated medication reminder and monitoring application called **Medi-Remind**. This system is designed to help users systematically organize, remember, and monitor their daily medication schedules. By using a digital medication-tracking system, users or authorized caregivers can record medication names, dosages, schedules, and other important medication instructions.

One of the key features of the system is the Medication Reminder, which automatically sends notifications when it is time for the user to take their medication. This feature is designed to reduce forgotten doses by providing timely reminders throughout the day. Another feature is the Medication Schedule, which allows users to organize their medications according to their prescribed dosage and time.

The system will also include Dose Confirmation, where users can mark a scheduled medication as “Taken” or “Missed.” This information will be recorded in the application and stored in the user’s medication history. By keeping a record of medication activities, users and caregivers can review whether scheduled medications were taken consistently.

Another important feature is Caregiver Notifications. When a scheduled dose is missed or remains unconfirmed, the system can notify a designated caregiver. This feature can provide additional support for older adults who may need assistance with medication management. Caregivers can also review the medication history to monitor the user's medication-taking activities.

Additional features include Refill Reminders, which can notify users when their medication supply may need to be replenished, and Medication History, which records previous taken and missed doses. These features can help users maintain a more organized medication routine and provide caregivers with useful information when monitoring medication adherence.

The system primarily targets older adults who regularly take medications, as well as their family members and caregivers. Healthcare professionals may also benefit from medication adherence records when information is properly authorized and shared. However, the application is intended as a support and reminder tool and does not replace professional medical advice, diagnosis, or instructions from healthcare providers.

The basic operation of the system will follow five stages: Add, Set, Remind, Confirm, and Monitor. First, the user or caregiver adds the medication details. Second, the dosage and schedule are set. Third, the application sends a reminder when the medication is due. Fourth, the user confirms whether the medication was taken or missed. Finally, the system records the activity and can notify the designated caregiver if a dose is missed.

The expected impact of this solution includes improved medication adherence, fewer forgotten doses, easier medication monitoring, better communication between older adults and caregivers, and greater independence for older adults. Overall, integrating medication scheduling, automated reminders, dose confirmation, medication history, refill reminders, and caregiver notifications offers a practical and sustainable approach to addressing medication non-adherence among older adults.

Conclusion

Medication non-adherence among older adults has become an important healthcare concern, with approximately 22% to 68% of older adults experiencing medication non-adherence (Ge, Heng, & Yap, 2023). Factors such as forgetfulness, multiple medications, complicated schedules, medication costs, and limited support can make it difficult for older adults to follow their medication instructions correctly. These

problems can negatively affect their health and may lead to poor disease control, treatment failure, and additional healthcare needs.

Addressing this problem requires not only awareness but also practical and accessible solutions. This proposal highlights the importance of implementing Medi-Remind, an automated medication reminder and monitoring application designed to support older adults and their caregivers. By organizing medication schedules, providing automated reminders, allowing users to confirm doses, recording medication history, providing refill reminders, and notifying caregivers about missed doses, the proposed system provides a structured approach to medication management.

Ultimately, Medi-Remind aims to help older adults maintain consistent medication-taking habits while allowing caregivers to provide additional assistance when necessary. By integrating technology into medication management, the proposed system can help reduce forgotten doses, improve monitoring, and promote greater independence among older adults. Overall, the system offers a practical technology-based approach to supporting medication adherence and improving daily medication management. The goal of Medi-Remind is simple: “The right medicine, at the right time, every time.”

SOURCES:

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